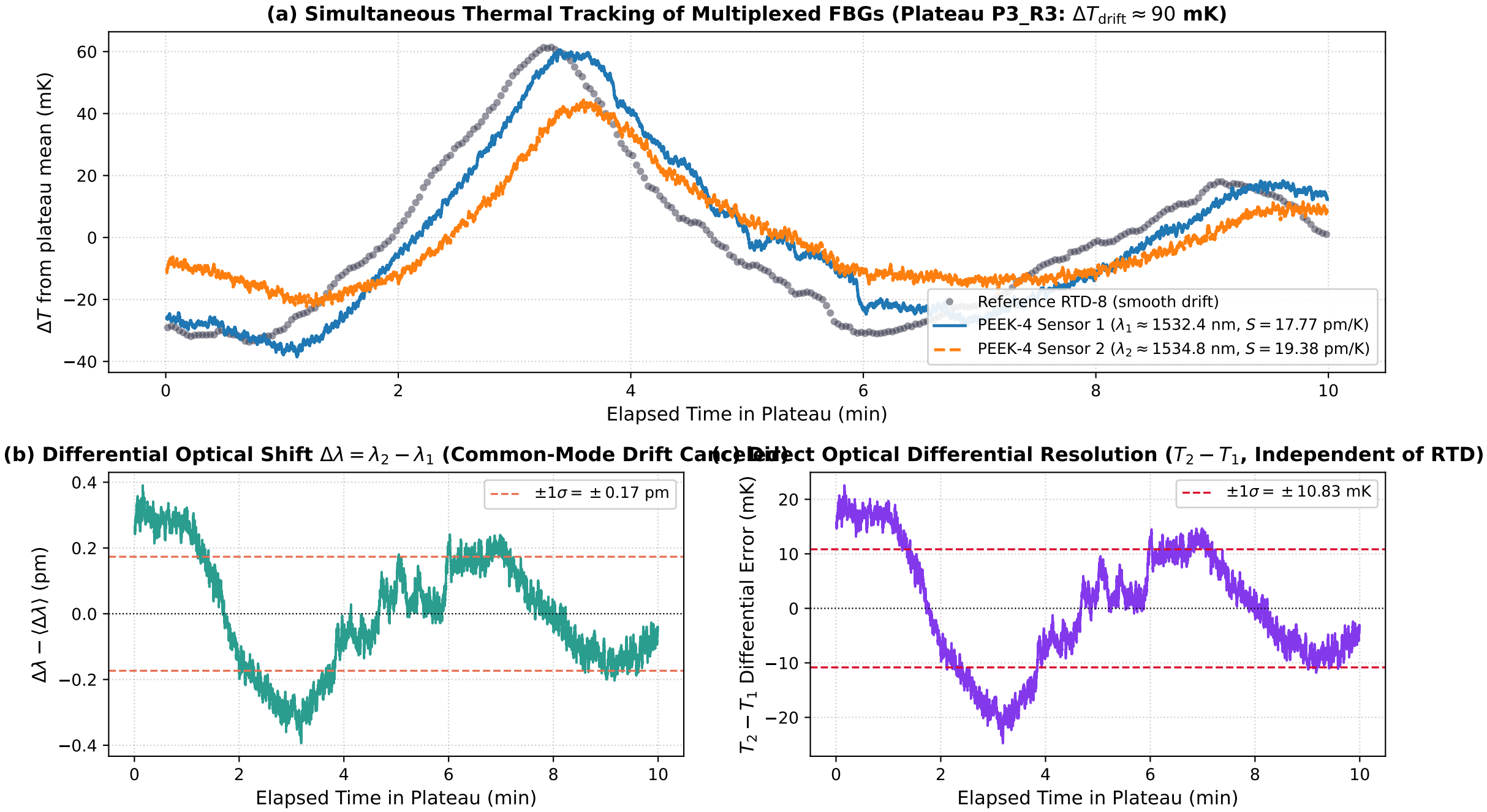

(a) Simultaneous Thermal Tracking of Multiplexed FBGs (Plateau P3_R3: $\Delta T_{drift} \approx 90$ mK)
ΔT from plateau mean (mK)
Elapsed Time in Plateau (min)
Reference RTD-8 (smooth drift)
PEEK-4 Sensor 1 ($\lambda_1 \approx 1532.4$ nm, $S = 17.77$ pm/K)
PEEK-4 Sensor 2 ($\lambda_2 \approx 1534.8$ nm, $S = 19.38$ pm/K)
(b) Differential Optical Shift $\Delta\lambda = \lambda_2 - \lambda_1$ (Common-Mode Drift Cancelled)
$\Delta\lambda - \langle\Delta\lambda\rangle$ (pm)
$\pm1\sigma = \pm 0.17$ pm
Elapsed Time in Plateau (min)
(c) Direct Optical Differential Resolution ($T_2 - T_1$, Independent of RTD)
$T_2 - T_1$ Differential Error (mK)
$\pm1\sigma = \pm 10.83$ mK
Elapsed Time in Plateau (min)